INCH-POUND

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MILITARY SPECIFICATION

SPAGHETTI WITH MEAT AND SAUCE, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostabilized spaghetti with meat and sauce in flexible pouches for use as a component of the Meal, Ready-to-Eat (MRE), Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Grades of Tomato Paste

(Copies are available from the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)
U.S. Standard of Identity for Enriched Macaroni Products
U.S. Standard of Identity for Parmesan Cheese

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) Cuts: Item No. 114 (shoulder, clod), Item No. 115 (chuck, square-cut, boneless), Item No. 128 (chuck, cross-cut, boneless), Item No. 133 (triangle, boneless), Item No. 161 (round, shank-off, boneless), Item No. 163 (round, shank-off, 3-way, boneless), Item No. 165 (round, rump, shank-off, boneless), Item No. 167A (round, knuckle, peeled), Item No. 169 (top round), Item No. 170A (bottom round, heel out), and Item No. 171B (outside round). All beef shall be certified by a USDA Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the IMPS Quality Assurance Provisions.

NOTE: Certified IMPS cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification, as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled as "Beef for further processing as specified in MIL-S-44205". The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

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b. Holding in the fresh-chilled state for not more than 4 days after initial certification is permitted, provided that the beef is maintained in the temperature range of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence.

d. Tempering/thawing of frozen beef shall comply with the following:

- microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by the USDA, is maintained
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
- regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen.

3.2.2 Tomatoes, canned, crushed or diced. Canned tomatoes shall be peeled, cored, mature, crushed or diced tomatoes. The use of safe and suitable firming and acidification ingredients and salt is permitted. The canned tomatoes shall have not less than 8.0 percent tomato soluble solids and shall possess a red flesh color, normal character, and a good distinct acid sweet tomato flavor and odor. The crushed or diced tomatoes shall be free of extraneous vegetable material and objectionable core material and skins (peel). The canned tomatoes shall be of the latest season's pack.

3.2.3 Tomato paste. The tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste and shall be of the latest season's pack. The tomato paste shall have not less than 26 percent tomato soluble solids.

3.2.4 Cheese, Parmesan, grated. The Parmesan cheese shall be grated and conform to the U.S. Standard of Identity for Parmesan Cheese. The cheese shall have been aged ten months or more. The grated cheese shall be free flowing and free from lumps that do not fall apart under light finger pressure.

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3.2.5 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.6 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked slurry prepared from the starch shall be bland with essentially no cereal or starch taste.

3.2.7 Onion, dehydrated, chopped. Dehydrated chopped onion shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Product.

3.2.8 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.9 Garlic powder. Garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Product.

3.2.10 Onion powder. Onion powder shall be Fancy grade of the Official Standard and Methods of the American Dehydrated Onion and Garlic Association for dehydrated onion and garlic products.

3.2.11 Oregano, ground. Ground oregano shall be derived from the dried leaves of Origanum vulgare L. and shall have a strong camphoraceous aroma and a pungent, slightly bitter flavor. The ground oregano shall contain not less than 2.0 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.12 Basil, ground. Ground basil shall be derived from the dried leaves of Ocimum basilicum L. and shall possess a sweet, anise-like odor and an aromatic, warm, slightly pungent flavor. The volatile oil content shall be not less than 0.4 mL per 100 grams of ground basil. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 35 sieve.

3.2.13 Pepper, red, ground. Ground red pepper shall be derived from red, ripe fruit of Capsicum frutescens L. and shall possess the characteristic yellowish-red to red color. The Scoville Pungency Value shall be not less than 30,000 units. The red pepper shall be uniformly ground to allow a minimum of 95 percent, by weight, to pass through a U.S. Standard No. 40 sieve, and not lest than 95 percent, by weight, to be retained on a U.S. Standard No. 60 sieve.

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3.2.14 Thyme, ground. Ground thyme shall be derived from the dried leaves and flowering tops of Thumus vulgaris L. and shall have a fragrant, aromatic odor and aromatic, minty flavor. The ground thyme shall contain not less than 0.8 mL of volatile oil per 100 grams of ground thyme and be of such size that not less than 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.15 Bay leaves, ground. Ground bay leaves shall be derived from dried leaves of Laurus nobilis L. and shall possess a pleasant, aromatic odor and pungent, mildly bitter flavor with a pale green to yellow green color. The volatile oil content shall be not less than 1.0 mL per 100 grams of ground bay leaves. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.16 Spaghetti, thin. The spaghetti shall be enriched, shall contain not less than 1.0 percent or more than 2.0 percent, by weight, of glycerol monostearate, and not less than 1.5 percent or more than 2.0 percent, by weight, of egg albumen. The spaghetti shall conform to the Standard of Identity for Enriched Macaroni Products except that the spaghetti shall be prepared from durum wheat semolina and shall have a 0.06 inch to 0.083 inch diameter. The thin spaghetti shall be of a solid cylindrical shape, shall be approximately 2 inches in length, and shall comply with MIL-STD-900.

3.2.17 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.18 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.2.19 Starch, waxy maize, modified, filling and processing aid. The starch shall be a white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hydrated at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial viscosity of the paste shall be retained under conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland and shall demonstrate easy dispersibility.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef meeting the requirements of 3.2.1 shall be further trimmed, if necessary, to assure compliance with finished product requirements. The beef shall be ground through a plate having holes of 3/8 inch (± 1/32 inch) in diameter. After grinding, the beef

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shall be held for not more than 24 hours at an internal temperature of 28°F to 40°F prior to the browning process (see 3.3.2).

NOTE: The following preparation procedures (3.3.2 through 3.3.4) were used in the development and production testing of this product. Alternative procedures may be used provided all finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified are still applicable.)

3.3.2 Sauce formulation and preparation. The sauce shall be formulated and prepared as follows:

Ingredient 1/	Percent by weight
Beef, ground	32.50
Water	30.63
Tomatoes, crushed or diced (8.0 percent solids) 1/	17.23
Tomato paste (26 percent solids) 1/	12.70
Parmesan cheese, grated	2.05
Salt 2/	1.25
Starch, food, modified, high opacity	1.25
Onion, chopped, dehydrated	1.20
Sugar, white, granulated	0.75
Garlic powder	0.17
Onion powder	0.10
Oregano, ground	0.08
Basil, ground	0.04
Pepper, red, ground	0.03
Thyme, ground	0.01
Bay leaves, ground	0.01

1/ If the specified tomato solids are not available, the tomato solids that are used must be adjusted to bring the solid level to that specified.

2/ The total amount of salt in the sauce formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

a. The starch and part of the water shall be mixed to form a slurry.

b. The ground beef shall be heated in a small portion of the formula water in a steam jacketed kettle unit with continuous stirring until the meat is fully cooked (browned, with no pink color visible).

c. The remaining water and the balance of the ingredients (other than the starch slurry) shall be uniformly blended with the cooked ground beef.

d. The starch slurry shall be added and the sauce heated with constant stirring until the sauce temperature reaches 180°F to 190°F and held in this temperature range for approximately five minutes.

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e. The volume of the sauce shall be adjusted with water prior to filling to compensate for evaporation loss during heating and holding.

f. The sauce shall be filled within four hours after cooking. If a hot fill method is used, the prepared sauce shall be held in the temperature range of 150°F to 180°F at all times prior to filling into the pouches. If a cold fill method is used, the prepared sauce shall be immediately cooled to 28°F to 40°F and in this temperature range at all times prior to filling into the pouches.

3.3.3 Spaghetti preparation. The spaghetti shall be prepared as follows:

a. The dry spaghetti shall be batch weighed.

b. The weighed spaghetti shall be blanched in boiling water until approximately double in weight while stirring constantly.

c. After blanching, the spaghetti shall be rinsed in cold water to a temperature less than 90°F.

d. The blanched spaghetti shall be reweighed.

e. If the weight of the blanched spaghetti is not at least double the dry weight of the spaghetti, the difference in weight shall be added as extra water to the sauce after adjustment of the cooked sauce volume for evaporation losses before filling.

f. The spaghetti shall be filled into pouches within two hours after completion of the blanching process.

3.3.4 Filling and processing aid, starch preparation. When a filling and processing aid is used, a solution (slurry) shall be prepared with the filling and processing aid starch and part of the sauce formulation water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of the product.

3.3.5 Product preparation. The cooked spaghetti and the meat sauce shall be combined in the following proportions, using either a single-stage or multi-stage filling operation:

Ingredient	Percent by weight
Meat sauce	83.0
Spaghetti	17.0

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 1 of MIL-P-44073.

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b. A hot fill system is permitted if compatible with the single-stage or multi-stage filling and sealing method used.

c. The temperature of the spaghetti and meat sauce at time of filling shall be as specified (see 3.3.2, and 3.3.3) for the applicable filling process used.

d. Each filled and sealed pouch shall be in the retort process within two hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch shall be not more than 0.35 ounces.

e. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. The average net weight shall be not less than 8.0 ounces.

g. No individual pouch shall contain less than 7.5 ounces of product.

h. The average drained weight of the meat and spaghetti shall be not less than 5.6 ounces.

i. No individual pouch shall contain less than 5.0 ounces of drained meat and spaghetti.

j. Meat texture shall not be dry, rubbery, or mushy.

k. Spaghetti texture shall not be mushy or tough.

l. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

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m. The average fat content of the finished product shall be not greater than 9.0 percent and no individual pouch shall have a fat content greater than 11.0 percent.

n. The salt content of the product in any individual pouch shall be not greater than 1.3 percent nor less than 0.5 percent.

o. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC International and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

4.5.5 Product and drained weight inspection.

4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermostabilization operation. The pouches shall be heated in 180°F to 190°F water for 10 minutes, opened, and inspected. The lot

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size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The product contents shall be inspected for the defects indicated in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	The total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch is more than 0.35 ounce
102	Presence of a bone piece measuring 0.3 inch or more in any dimension
103	Texture of the meat is dry, rubbery, or mushy
104	Texture of the spaghetti is mushy or tough

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TABLE II. Product defects (cont'd) 1/ 2/

Category	Defect
Major	
105	Sauce consistency is lumpy or is excessively thick or thin
106	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, or metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermostabilization operation. The pouches shall be heated in 180°F to 190°F water for approximately 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table III. The drained weight shall be determined and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for drained weight inspection 1/ 2/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined, and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Drained weight defect 1/

Category	Defect
Major	
101	Drained weight of meat and spaghetti in a pouch is less than 5.0 ounces 2/

1/ To determine drained weight, the pouch contents shall be poured into a flat-bottom container. A minimum of three times the pouch's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquefy rendered fat and remove the sauce. The contents shall then be poured into a U.S. Standard No. 20 sieve in a manner that will distribute the product over the sieve. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

2/ If the sample average drained weight of meat and spaghetti is less than 5.6 ounces, the lot shall be rejected.

4.5.6 Fat and salt content testing. Eight filled and sealed pouches shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, methods 976.21, 985.15, or 960.39, and for salt content in accordance with the Official Methods of Analysis of AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened pouches shall be gently warmed in a water bath to melt fat adhering to the inside of the pouches. The pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

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5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Combat field feeding
Food processing
Operational ration
Pasta
Pouch food
Shelf stable

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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a visor style, having a circular top crown, two ... 3.3 Design. The cap is of

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8940-0750)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-44205B	1993 September 1

3. DOCUMENT TITLE

SPAGHETTI WITH MEAT AND SAUCE, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WRE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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identification/instruction label conforming to type VI, class 13 of bbb-l-20.